Current Equation Verification for the Raman-Suppression Codebase

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This document is the current verification backbone for the research-note series. It checks the equations that matter most for the active codebase: the interaction-picture forward model, Raman-band objective, adjoint phase gradient, regularized dB objective, reduced-basis chain rule, Hessian and trust-region interpretation, multimode gradients, and multivariable chain-rule paths. The goal is not to replace every research note with one huge derivation. The goal is to give each note a compact, auditable place to point when a reader asks, "Where did this equation come from, where is it in code, and how was it checked?" Some items are verified by existing tests and saved artifacts. A few heavyweight reruns are still pending and are explicitly marked as such.

Claim Status

Status: current verification map; burst reruns pending
The derivations and source-path audit below reflect the current codebase as of 2026-04-26. The single-mode phase-gradient path, regularized objective convention, reduced-basis chain rule, finite-difference HVP interpretation, and MMF mode-coordinate finite-difference policy are ready to cite with the caveats shown. The full production verification suite and the multivariable gradient smoke suite still need fresh burst reruns before the whole note series should be called publication-grade.

1 How To Read This Document

Each equation is checked through the same stack:

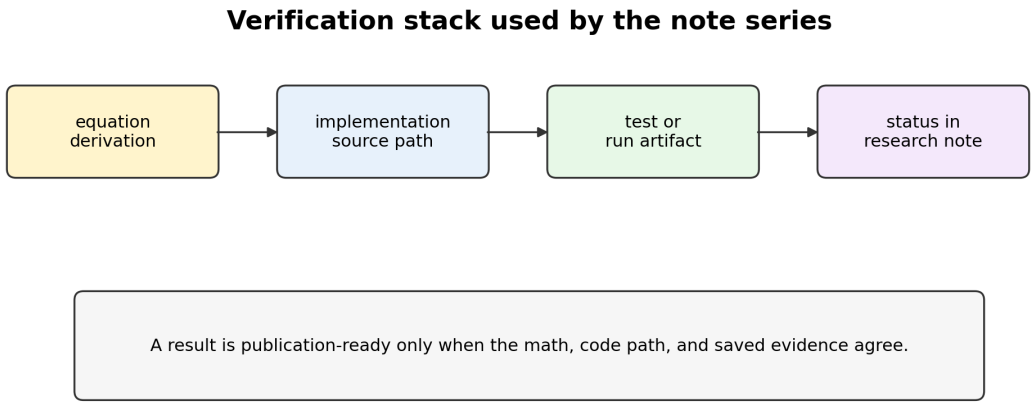

Figure 1: Verification stack. A mathematical claim is only useful for the note series when the derivation, source path, and test or run artifact all agree.

Verified means the equation is implemented and has an appropriate finite-difference, Taylor, identity, or regression check.

Verified with caveat means the main equation is usable, but the note must state an important scope limit.

Finite-difference by design means the project intentionally does not claim an analytic derivative for that block.

Pending rerun means the code path has evidence, but a current publication-grade run has not yet been refreshed.

Intuition Check

The point is not to make every reader redo all the math. The point is to make the logic traceable: equation, code, check, and conclusion.

2 Forward Fiber Model

The propagated spectral field in the lab frame is $\hat{u}(z,\omega)$. Here ω denotes the absolute angular-frequency grid used by the code. If a derivation instead uses the offset $\Omega = \omega - \omega_0$, the self-steepening factor below is $1 + \Omega/\omega_0$. The generalized nonlinear Schrodinger equation used by the project is the standard nonlinear-fiber model with dispersion, Kerr nonlinearity, delayed Raman response, and self-steepening [1, 2]. In compact form:

$$\frac{\partial \hat{u}}{\partial z} = iD(\omega)\hat{u} + i\frac{\omega}{\omega_0}\mathcal{F}\left[\gamma(1-f_R)|u|^2 u + \gamma f_R (h_R * |u|^2)u\right].$$

The implementation stores the delayed Raman fraction inside the discrete Raman-response array and stores the instantaneous fraction separately as $1-f_R$. This is algebraically the same convention as the equation above if h_R is interpreted as a unit-normalized Raman response.

The code integrates the interaction-picture field

$$\tilde{u}(z,\omega) = e^{-iD(\omega)z}\hat{u}(z,\omega).$$

Differentiating gives

$$\frac{\partial \tilde{u}}{\partial z} = ie^{-iD(\omega)z}\frac{\omega}{\omega_0}\mathcal{F}\left[\gamma(1-f_R)|u|^2u + \gamma f_R(h_R * |u|^2)u\right],$$

where the nonlinear term is evaluated from the lab-frame field $\hat{u} = e^{iDz}\tilde{u}$.

Convention check. With the code's interaction-picture definition, the lab-frame linear term is $+iD(\omega)\hat{u}$. Changing the Fourier sign convention or redefining D can move the minus signs around, so verification must compare the complete pair: transform convention plus dispersion definition plus self-steepening factor.

Implementation path	What to check
`src/simulation/simulate_disp_mmf.jl`	`disp_mmf!` applies e^{iDz}, transforms to time, builds Kerr and Raman terms, applies self-steepening, and returns to the interaction picture.
`scripts/lib/common.jl`	Problem setup builds grids, Raman response, fiber dictionaries, and the Raman-band mask.

Verification status: verified with caveat. The equation matches the standard GNLSE form and the implementation is source-audited. The old production verification script includes soliton preservation, photon-number conservation, and direct band-cost checks. A fresh burst rerun of that script is still required before calling this fully publication-refreshed.

Intuition Check

The interaction picture removes the fast linear dispersion spin from the ODE. The solver then spends most of its effort on the nonlinear part that actually changes the pulse shape.

3 Raman-Band Cost And Terminal Condition

The physical cost is the fraction of output spectral energy in the Raman band:

$$J_{\text{phys}}(\phi) = \frac{E_{\mathcal{B}_{\text{Raman}}}}{E_{\text{tot}}} = \frac{\sum_{\omega_k\in\mathcal{B}_{\text{Raman}}}|u_{\omega_k}(L;\phi)|^2}{\sum_k |u_{\omega_k}(L;\phi)|^2}.$$

Let χ_k be 1 inside the Raman band and 0 outside. With $J = E_{\mathcal{B}_{\text{Raman}}}/E_{\text{tot}}$, the derivative with respect to the output field conjugate is

$$\frac{\partial J}{\partial u_k^*(L)} = \frac{u_k(L)(\chi_k - J)}{E_{\text{tot}}}.$$

This is the terminal condition for the adjoint solve.

Implementation path	What to check
`scripts/lib/common.jl`	`spectral_band_cost` computes $E_{\mathcal{B}_{\mathrm{Raman}}}$, E_{tot}, J, and $u(\chi - J)/E_{\mathrm{tot}}$.
`scripts/research/analysis/verification.jl`	Directly compares `spectral_band_cost` against a separate energy-ratio computation.

Verification status: verified; production rerun pending. The quotient derivative is analytically correct. The verification script contains a machine-precision direct-cost cross-check, but the current publication pass should rerun it on burst.

Intuition Check

The terminal adjoint says: increase sensitivity where energy is inside the bad band, and decrease sensitivity elsewhere in proportion to the total-energy normalization.

4 Adjoint Phase Gradient

The input phase mask is

$$u_\omega(0;\phi) = u_{\omega,0} e^{i\phi(\omega)}.$$

The adjoint solve returns the sensitivity of J to the input field. For a real-valued objective,

$$dJ = 2\,\mathrm{Re}\langle \lambda_0, du_\omega(0)\rangle.$$

Since

$$du_\omega(0) = i u_\omega(0;\phi)\, d\phi,$$

the phase gradient is

$$\boxed{\frac{dJ}{d\phi(\omega)} = 2\,\mathrm{Re}\left[\lambda_0(\omega)^*\, i\, u_\omega(0;\phi)\right].}$$

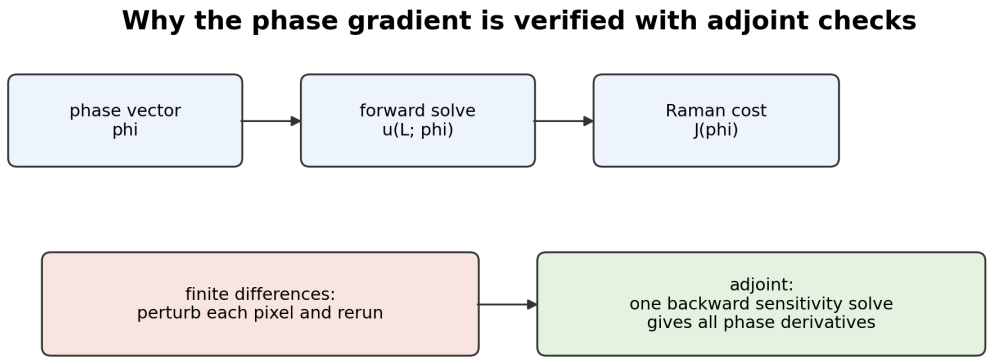

Figure 2: Adjoint-gradient verification logic. The adjoint replaces a pixel-by-pixel finite-difference gradient with one backward sensitivity solve, but it must still be checked against finite-difference or Taylor behavior.

Implementation path	What to check
`src/simulation/sensitivity_disp_mmf.jl`	`adjoint_disp_mmf!` implements the backward sensitivity RHS using the stored forward solution.
`scripts/lib/raman_optimization.jl`	`cost_and_gradient` applies the input phase chain rule $2\,\mathrm{Re}[\lambda_0^* i u_0]$.
`scripts/research/analysis/verification.jl`	Taylor remainder should scale like $O(\epsilon^2)$.

Verification status: verified with caveat. The phase chain rule is straightforward and the source code matches it. The full adjoint RHS is more complex because Kerr and Raman terms contain u, u^*, convolution paths, and mode tensors. The project verifies the full assembled gradient by finite-difference/Taylor behavior rather than asking a reader to trust every RHS subterm by inspection alone.

Intuition Check

The adjoint is the "all knobs at once" derivative. It tells the optimizer which phase pixels should move up or down without rerunning the fiber once per pixel.

5 Regularized dB Objective

The optimizer may use a regularized linear surface

$$J_{\mathrm{lin}}(\phi) = J_{\mathrm{phys}}(\phi) + \lambda_{\mathrm{gdd}} R_{\mathrm{gdd}}(\phi) + \lambda_{\mathrm{edge}} R_{\mathrm{edge}}(\phi).$$

If the run uses a dB objective, the scalar surface is

$$J_{\mathrm{surf}}(\phi) = 10 \log_{10} J_{\mathrm{lin}}(\phi).$$

The gradient must be scaled by the chain rule:

$$\nabla J_{\mathrm{surf}} = \frac{10}{\ln 10} \frac{1}{J_{\mathrm{lin}}} \nabla J_{\mathrm{lin}}.$$

This is an important repaired convention: regularizers are added first, and the dB transform is applied to the full regularized scalar.

The GDD regularizer uses the second-difference stencil

$$R_{\mathrm{gdd}}(\phi) = \sum_{i=2}^{N_t-1} \frac{(\phi_{i+1} - 2\phi_i + \phi_{i-1})^2}{\Delta\omega^3}.$$

In matrix form, if D_2 is the nonperiodic second-difference matrix with row stencil $(1, -2, 1)$, then

$$R_{\mathrm{gdd}}(\phi) = \frac{1}{\Delta\omega^3} \|D_2\phi\|_2^2, \qquad \nabla R_{\mathrm{gdd}} = \frac{2}{\Delta\omega^3} D_2^T D_2 \phi.$$

The strict interior stencil is proportional to

$$\phi_{j-2} - 4\phi_{j-1} + 6\phi_j - 4\phi_{j+1} + \phi_{j+2}.$$

The first two and last two samples use the corresponding truncated rows from $D_2^T D_2$, not a periodic wraparound.

The boundary regularizer measures temporal edge fraction of the shaped input pulse:

$$R_{\mathrm{edge}}(\phi) = \frac{\sum_{t_j \in \mathrm{edge}} |u(t_j, 0; \phi)|^2}{\sum_j |u(t_j, 0; \phi)|^2}.$$

For phase-only optimization, the denominator is constant under the FFT by Parseval's theorem because phase changes preserve spectral amplitudes. For amplitude-enabled optimization, the denominator is not constant and the full quotient rule is required. This is the exact distinction that the repaired multivariable boundary-amplitude gradient checks.

Implementation path	**What to check**
`scripts/lib/regularizers.jl`	Adds GDD and boundary regularizer scalar contributions and gradients.
`scripts/lib/objective_surface.jl`	Records whether the scalar surface is linear or dB and whether regularizers are chained into the surface.
numerics regression tests	Regression coverage for the repaired objective-scale convention.

Verification status: verified. This is the current authoritative objective convention for single-mode phase optimization.

Intuition Check

Do not compare two results unless you know what surface they optimized. A dB objective, a linear objective, and a regularized objective can all report a Raman number, but they did not ask the optimizer the same question.

6 Coordinate Maps And Chain Rules

Several research paths optimize variables that are not the raw full-grid phase. The verification rule is always the same: map optimizer coordinates to physical controls, then chain-rule gradients back to the optimizer coordinates.

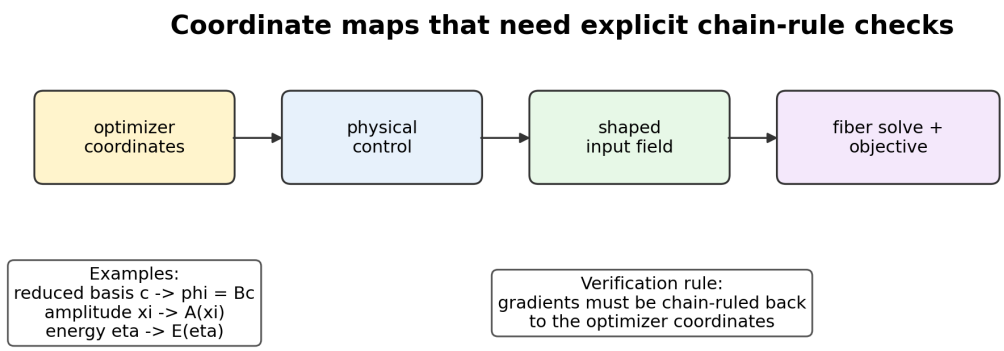

Figure 3: Coordinate-map verification rule. Reduced-basis, amplitude, energy, and mode-coordinate controls all need explicit chain-rule or finite-difference checks in the coordinates the optimizer actually uses.

6.1 Reduced Basis

Reduced-basis phase optimization uses

$$\phi = Bc,$$

where columns of B are allowed phase shapes and c is the optimizer coordinate vector. The gradient must satisfy

$$\nabla_c J_{\text{surf}} = B^T \nabla_\phi J_{\text{surf}}.$$

Implementation path: reduced-basis and continuation code under `scripts/research/phases/` plus the reduced-basis and basis-extension regression tests.

Verification status: verified for the active reduced-basis lane. The final public reduced-basis note still needs a provenance table connecting reported numbers to saved artifacts.

6.2 Multivariable Phase, Amplitude, And Energy

The multivariable path can optimize phase, spectral amplitude, and scalar energy. The same forward-adjoint sensitivity is chained through each variable. For example, an amplitude parameterization $A(\xi)$ needs

$$\frac{dJ}{d\xi} = \frac{dJ}{dA}\frac{dA}{d\xi}.$$

The repaired boundary-amplitude gradient uses the quotient rule because edge fraction depends on both edge energy and total input energy.

Implementation path: `scripts/research/multivar/multivar_optimization.jl`.

Verification status: pending rerun. A local algebra check matched central finite differences for the repaired boundary-amplitude derivative. The burst multivariable smoke suite still needs to be rerun before final amplitude-enabled claims are trusted.

7 Hessian And Trust-Region Diagnostics

The project uses Hessian-vector products and trust-region methods as diagnostics and candidate optimizers. The key verification point is negative: the current HVP paths should be described as finite-difference Hessian diagnostics unless a note explicitly proves an analytic second-adjoint path.

If $F(\phi) = 10\log_{10} J_{\text{lin}}(\phi)$, then

$$\nabla^2 F = \frac{10}{\ln 10}\left(\frac{1}{J_{\text{lin}}}\nabla^2 J_{\text{lin}} - \frac{1}{J_{\text{lin}}^2}\nabla J_{\text{lin}}\nabla J_{\text{lin}}^T\right).$$

So a linear-cost Hessian is not automatically the Hessian of the regularized dB optimizer surface.

Implementation paths: HVP regression tests, `test/trust_region/`, and trust-region code under `scripts/research/trust_region/`.

Verification status: verified with caveat. HVPs are useful when the scalar surface is named. Public notes must not call them analytic Hessians unless that is actually the path being tested.

Intuition Check

Curvature depends on the surface. If we change the cost scale from linear to dB, we changed the hill the optimizer is walking on.

8 Multimode Derivatives

For shared-phase multimode optimization, the phase is common across modes but the cost depends on all propagated modal fields. The phase gradient sums the modewise sensitivities:

$$\frac{dJ}{d\phi(\omega)} = \sum_m 2\,\mathrm{Re}\left[\lambda_{0,m}(\omega)^*\, i\, u_{0,m}(\omega)\right].$$

For phase-plus-mode-coordinate work, the mode-coordinate block is currently finite-differenced by design. That is acceptable because the block is small relative to the phase grid, and a previous analytic complex chain did not pass preflight.

Implementation paths:

The MMF research-driver directory contains the Raman and joint-optimization drivers. The MMF regression tests check the derivative path.

Verification status: shared-phase verified; mode-coordinate block finite-difference by design. A clean burst preflight for the mode-coordinate finite-difference block reported zero relative error in the saved summary, but the MMF regression test should still be rerun for the final verification pass.

9 Current Verification Matrix

Path	Equation status	Evidence	Public status
Forward interaction-picture GNLSE	analytic model from nonlinear fiber optics	source audit; soliton/photon verification script exists	verified with fresh burst rerun pending
Raman-band cost	quotient derivative analytic	direct cost cross-check exists	verified, rerun pending
Single-mode phase gradient	analytic adjoint chain rule	Taylor and FD checks	verified
GDD regularizer	analytic discrete stencil	regularized gradient tests	verified
Boundary phase regularizer	analytic phase derivative	regularized gradient tests	verified
Reduced basis	linear map $\phi = Bc$	basis tests and extension tests	verified, provenance table still needed
Sharpness objective	mixed objective / robustness diagnostic	sharpness tests	verified with scalar-surface caveats
HVP / trust-region	finite-difference HVP diagnostics	HVP and trust-region tests	verified with caveat
MMF shared phase	analytic mode-summed chain rule	MMF FD tests	verified, rerun pending
MMF mode coordinates	not analytic by current policy	finite-difference preflight	finite-difference by design
Multivariable amplitude boundary term	analytic quotient-rule derivative	local FD check	pending burst smoke rerun

10 Required Reruns Before Final Freeze

These should run on the burst machine through the approved heavy-job wrapper:

1. `julia -t auto -project=. scripts/research/analysis/verification.jl`
2. `julia -t auto -project=. scripts/dev/smoke/test_multivar_gradients.jl`
3. Run the active MMF regression entry point from the test manifest, using Julia threading.

The cost-audit missing matrix should only be rerun if the project decides that the cost-audit note needs completed matrix coverage rather than an honest "matrix incomplete" status.

11 What This Means For The Research Notes

Baseline and reduced-basis notes can cite the adjoint phase-gradient, regularized objective, and reduced-basis chain-rule sections here, but still need result-provenance tables.

Sharpness and trust-region notes must name the scalar surface for every HVP or Hessian claim.

Multimode and multivariable notes should remain provisional until the specified reruns and result-regeneration decisions are complete.

Cost/numerics note should point here once the burst reruns are attached, so it does not need to carry every derivation itself.

12 Sources

References

[1] G. P. Agrawal. *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019. https://doi.org/10.1016/C2018-0-01168-8

[2] J. Hult. "A fourth-order Runge-Kutta in the interaction picture method for simulating supercontinuum generation in optical fibers." *Journal of Lightwave Technology*, 25(12), 3770–3775, 2007. https://doi.org/10.1109/JLT.2007.909373

[3] J. Nocedal and S. J. Wright. *Numerical Optimization*, 2nd ed. Springer, 2006. https://doi.org/10.1007/978-0-387-40065-5

[4] M. Frigo and S. G. Johnson. "The design and implementation of FFTW3." *Proceedings of the IEEE*, 93(2), 216–231, 2005. https://doi.org/10.1109/JPROC.2004.840301